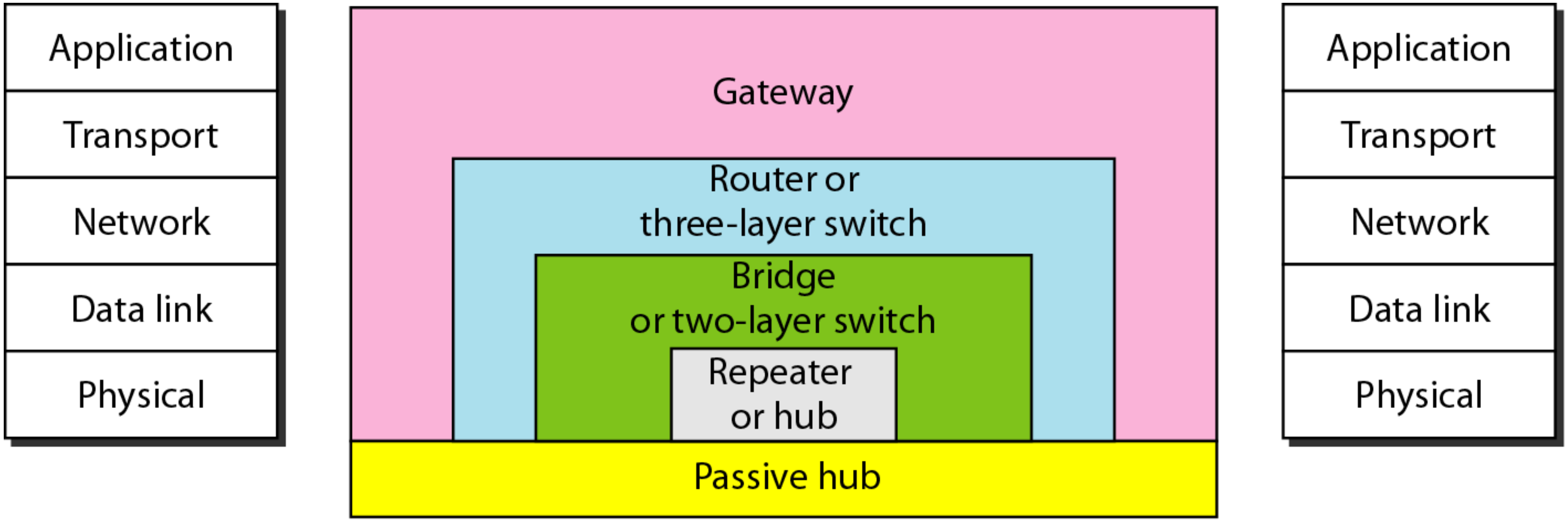
Connecting Devices

HUB, SWITCH, ROUTER, FIREWALL, GATEWAY, NIC, REPEATERS

Introduction

- Connecting devices are used to connect multiple computers.
- Connecting devices are divided into 5 different categories based on the layer in which they operate in a network.
- The 5 categories contain devices can be defined as:
    - those operate below the physical layer (passive hub)
    - those operate at the physical layer (a repeater or an active hub)
    - those operate at the physical & data link layers (a bridge or a 2-layer switch)
    - those operate at the physical, data link & network layers (a router or a 3-layer switch)
    - those operate at all 5 layers (a gateway)



Passive Hub (Historical Device)

- A passive hub is just a connector.
- It connects the wire coming from different branches.
- Location in the Internet model is below the physical layer.

Passive hubs are obsolete and are included only for understanding the evolution of networking devices. Modern LANs use Ethernet switches instead.

Hubs

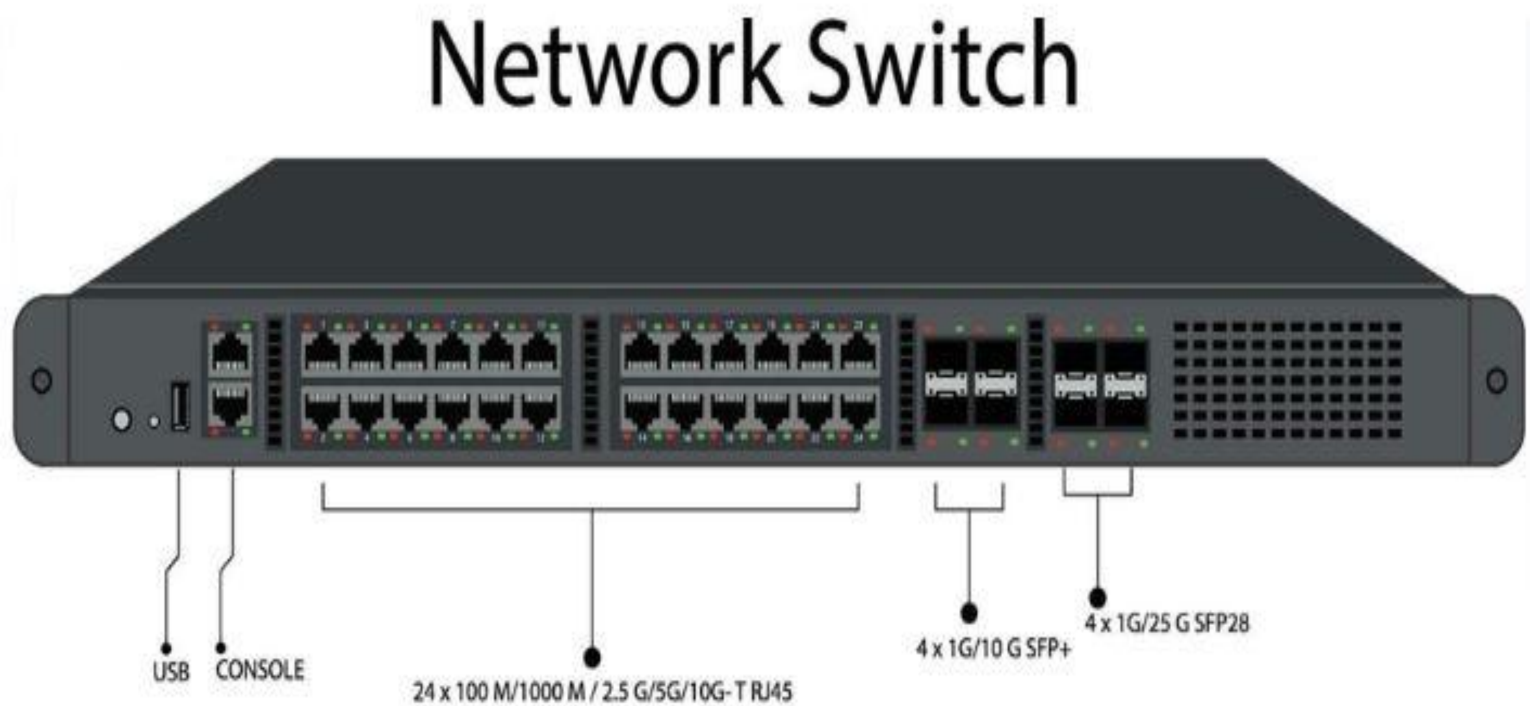
- Hubs are intermediary devices that usually operate at the physical layer (Layer I) of the OSI model that connect multiple devices on a local network.
- Hubs function by broadcasting data to all the devices that are connected to it, regardless of their destination.
- Hubs can help in simplifying the wiring and installation of a LAN by providing a central point of connection. Hubs are capable of creating a single collision domain and a single broadcast domain.

Hubs

Switch

- Switches serve as networking devices that function at the data link layer (layer 2) of the OSI model.
- The main purpose of switches is to connect end devices within a network and allow the forwarding of data packets utilizing their MAC addresses.

Switch


Network Switch
USB
CONSOLE
24 x 100 M/1000 M / 2.5 G/5G/10G-T RJ45
4 x 1G/10 G SFP+
4 x 1G/25 G SFP28

Router

- Routers are also networking devices that operate at the network layer (layer 3) of the OSI model.
- The main function of the router is to connect networks and allow the forwarding of data packets based on their respective IP addresses.

Firewall

- A firewall is an intermediary device that operates at various layers of the OSI Model depending on its type.
- It can be either hardware or software, and its main purpose is to protect the network from any unauthorized access or malicious attacks by enforcing policies or rules.

Gateway

- A gateway can operate at multiple OSI layers depending on the application because it performs protocol translation between different networks or architectures..
- They do so by translating or converting data formats or protocols between different networks.

NIC

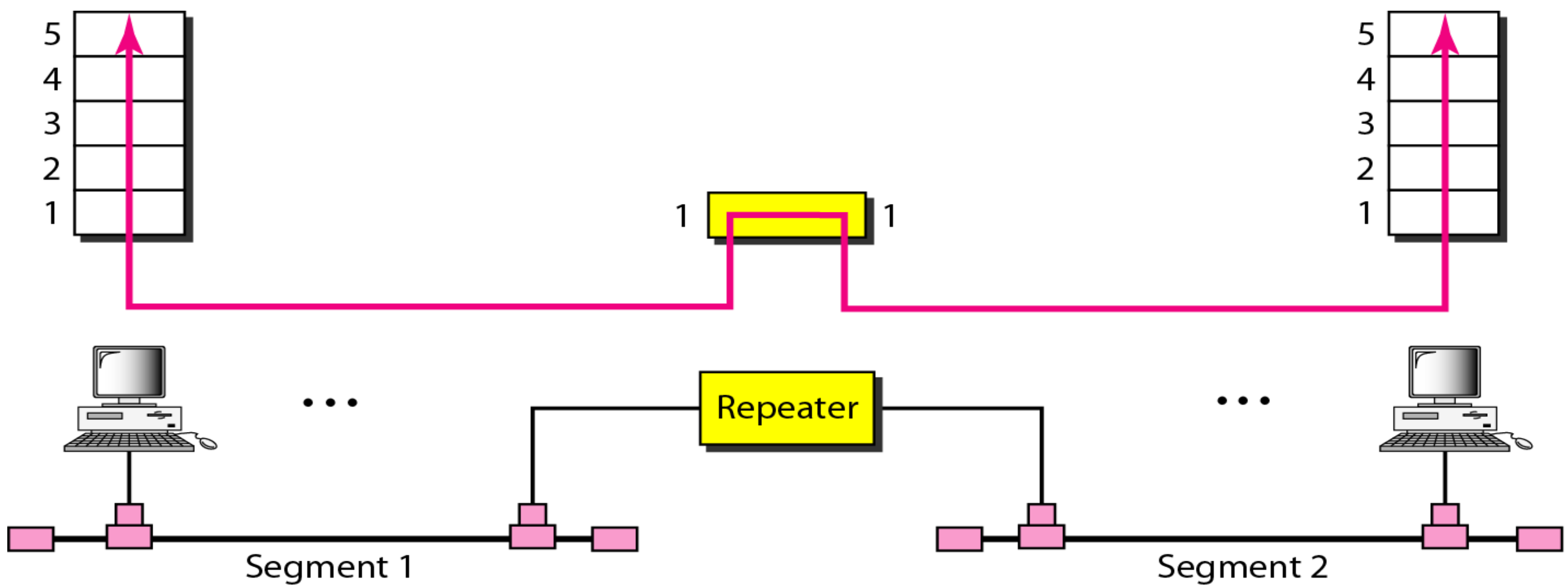
- NIC stands for Network Interface Card and operates at the data link layer (Layer II) of the OSI model.
- It is a hardware component that is responsible for enabling a device to connect to a network. A NIC also has one or more ports that allow different types of cables or wireless signals to connect to the network.

Repeater

- Repeaters are sometimes also known as “Signal Booster”. The main function of the repeater is to regenerate an incoming signal from the sender before retransmitting it to the receiver.

- The layer at which it operates is the physical layer of the OSI model.

- Repeater mainly used to extend the physical length of a LANs.
- Connects 2 segments of the same LAN.
- It cannot be used to connect LAN of two different protocol.



Device	Use	Example
Hub	Used to connect multiple computers in a very small network. Rarely used today because it sends data to all devices, causing collisions.	A small computer lab in the 1990s where 8 PCs were connected using a hub. Today, almost completely replaced by switches.
Repeater	Extends the transmission distance by regenerating weak signals.	Wi-Fi range extenders in homes, optical fiber repeaters in undersea internet cables, mobile signal boosters.
NIC (Network Interface Card)	Provides the hardware interface that allows a device to connect to a network. Every network-enabled device has one.	Laptop Ethernet port, Wi-Fi adapter in smartphones, network card in servers, smart TVs, printers.
Switch	Connects devices within the same Local Area Network (LAN) and intelligently forwards data only to the intended device.	Office networks, college laboratories, banks, hospitals, data centers where hundreds of computers communicate efficiently.

Device	Use	Example
Router	Connects different networks and forwards packets using IP addresses. Enables Internet access.	Home Wi-Fi router connecting your home network to the Internet, routers used by ISPs, enterprise branch offices connected through WAN.
Gateway	Connects networks using different protocols or architectures by translating between them.	IoT gateway connecting ZigBee sensors to an IP network, VoIP gateway connecting traditional telephone systems to Internet calling, cloud gateways.
Firewall	Protects a network by allowing or blocking traffic based on security rules.	Banks blocking unauthorized access, company offices preventing hackers, cloud firewalls protecting web applications, Windows Firewall protecting personal computers.

Example: Home Network

Router (Wi-Fi): Is this the router or access point?

- In a typical home network, the device labelled **Router (Wi-Fi)** is actually a **wireless router**, which combines multiple networking devices into one.
- It includes:
  - Router (Layer 3) – Connects your home network to the Internet (ISP).
  - Wireless Access Point (Layer 2) – Provides Wi-Fi connectivity to wireless devices.
  - Switch (Layer 2) – Usually has 4 Ethernet LAN ports for wired devices.
  - NAT & DHCP Server – Assigns IP addresses and translates private IPs to the public Internet.
  - Basic Firewall – Protects the home network from unauthorized incoming traffic.

- Home: One wireless router performs the jobs of a router, switch, access point, DHCP server, NAT device, and firewall.
- Enterprise: These are separate dedicated devices for better performance, scalability, and management.

Device	Layer (OSI)	Main Function	Still Common Today?
Hub	Layer 1	Broadcasts data to all ports	Rarely used
Repeater	Layer 1	Regenerates weak signals	Yes
NIC	Layer 2 (Hardware interface)	Connects a device to a network	Every device
Switch	Layer 2	Connects devices within a LAN	Very common
Router	Layer 3	Connects different networks	Essential
Gateway	Layers 4–7 (typically)	Protocol conversion and interoperability	Common in IoT, cloud, telecom
Firewall	Layers 3–7 (depending on type)	Network security and traffic filtering	Essential

Question?

Case Study: Network Design for State Bank of India (SBI) – New Smart Branch

- State Bank of India (SBI) is opening a three-storey smart branch in Mumbai with modern banking infrastructure. As a Network Engineer, you have been assigned to design the branch network.
- Branch Infrastructure:50 Employee Computers , 8 Cash Counters, 5 ATM Machines, 10 IP CCTV Cameras, 6 Wi-Fi Access Points, 3 Network Printers, 1 Central Server Room, Internet Connection from ISP

- Business Requirements
    - Employees should access the Internet securely.
    - Customers should use Guest Wi-Fi only.
    - ATMs should communicate only with the SBI Head Office.
    - Internal servers must be protected from hackers and cyberattacks.
    - The network should continue operating even if one cable fails.
    - All three floors should have high-speed, reliable connectivity.

Task

- Design a suitable network for the SBI branch by selecting the appropriate networking devices:
  - Hub, Switch, Router, Firewall, Gateway, NIC, and Repeater.
  - Explain:
    - Where each device should be installed.
    - The function of each device.
    - Why it is the best choice for this banking network.
    - Why some devices (e.g., Hub) are not suitable for modern enterprise networks.

Solution: Identify the Networking Devices Required

Device	Quantity	Purpose
Router	1	Connect branch network to ISP and Head Office
Firewall	1	Secure the entire banking network
Core Switch	1	Connect all access switches
Access Switch	3 (one per floor)	Connect employee PCs, ATMs, printers, APs, cameras
Gateway	1	Connect banking applications using different protocols
Repeater/Fiber Repeater	As required	Extend communication beyond cable distance
NIC	One in every device	Enables network communication
Hub	Not Used	Obsolete and insecure

Function of Each Device

NIC (Network Interface Card)

- Definition: A NIC is a hardware component installed in every network-enabled device that allows it to communicate over a network.
- Installed In
  - Desktop Computers
  - ATM Machines
  - Servers
  - Printers
  - CCTV Cameras
  - Wi-Fi Access Points

- The NIC converts digital data into electrical or wireless signals for transmission.
- Why Required
  - Without a NIC, no device can join the network.

Core Switch

- Definition: A Core Switch acts as the central backbone of the enterprise network and connects all access switches.
- Installed in Server Room.
- Functions
  - Connect all floors
  - High-speed switching
  - VLAN routing
  - Connect server room
- Why Core Switch?
  - Provides
  - High bandwidth
  - Low latency
  - Scalability
- Example: Cisco Catalyst 9500

Access Switch

- An Access switch connects multiple devices in a Local Area Network (LAN) and forwards data only to the intended destination using MAC addresses.
- Installed
  - One managed switch on each floor.
- Connected Devices
  - Computers
  - Printers
  - ATMs
  - IP Cameras
  - Wi-Fi Access Points

- Working
  - Suppose Computer A wants to send data to Printer 1.
  - The switch checks its MAC Address Table.
  - Unlike a hub, the frame is sent only to the printer.
- Advantages
  - No collisions
  - High speed
  - Secure communication
  - Efficient bandwidth usage
- Example: Cisco Catalyst 9200 Series

Router

- Definition: A Router is a Layer-3 networking device that connects multiple networks and forwards packets based on IP addresses.
- Installed in Server Room between ISP & Firewall.
- Functions
  - Connect SBI Branch to Internet
  - Connect Branch to SBI Head Office
  - Performs inter-VLAN routing when configured as a Layer 3 router or when used with a Layer 3 switch.
  - Provide default gateway

- Why Router?
  - Supports static and dynamic routing
  - Secure WAN connectivity
  - VPN connectivity with Head Office
  - High performance
- Example: Cisco ISR 4331 or Cisco Catalyst 8300.

Firewall

- Definition: A Firewall monitors and filters incoming and outgoing network traffic based on predefined security rules.
- Installed Between Router & Core Switch
- Functions
  - Prevent cyber attacks
  - Block unauthorized access
  - Protect internal servers
  - Secure Internet access

- **Why Firewall?**
  - SBI handles
  - Customer accounts
  - Financial transactions
  - ATM communication
  - These require maximum security.
- **Example**
  - Cisco Firepower
  - Palo Alto
  - Fortinet FortiGate

Gateway

- Definition: A Gateway is the device that enables communication between networks using different protocols or between the local network and external networks.
- ***Installation: Router acts as Default Gateway***
- Functions
  - Connect LAN to Internet
  - Connect Branch to SBI Head Office
  - Route packets outside local network

Why Required?

- Suppose Employee PC wants to access

www.onlinesbi.com

- The PC sends the packet to Default Gateway → Router → Internet.

Fiber Repeater / Media Converter

- **Definition:** A **Repeater** regenerates weak signals to extend communication distance.
- **Installed** Between Floors if Fiber length exceeds standard limits.
- **Functions**
  - Extend signal
  - Reduce attenuation
  - Maintain communication quality
- **Why Required?**
- SBI building has 3 Floors. Large cable lengths are required to cover, so it is hard to implement. Hence, repeaters are required.

Fiber repeaters maintain high-speed communication.

Modern enterprise networks generally use optical transceivers and switches for long-distance Fiber communication. Dedicated repeaters are mainly used in specialized long-distance links such as undersea cables and telecom backbone networks.

Hub

- Definition: A Hub broadcasts every received frame to all connected devices.
- Why Not Used?

Problem	Impact
Broadcasts every frame	Low Security
Creates collisions	Poor Performance
Half Duplex	Slow
Easy packet sniffing	Banking data risk

- **Banking Industry**
  - Modern banks never use Hubs. They always preferred Ethernet Switches.

VLAN Design (For a Secure Network Design)

Why VLANs?

- ATM traffic isolated
- Guest Wi-Fi cannot access bank servers
- Better security
- Easier management

VLAN	Devices
VLAN 10	Employees
VLAN 20	Cash Counters
VLAN 30	ATM Machines
VLAN 40	CCTV Cameras
VLAN 50	Guest Wi-Fi
VLAN 60	Servers

Network Security Measures

- Firewall Rules
- VLAN Segmentation
- WPA3 for Wi-Fi
- VPN to SBI Head Office
- Access Control Lists (ACLs)
- Intrusion Detection/Prevention
- Regular Backups

Component	Standard
Ethernet	IEEE 802.3
Wi-Fi	IEEE 802.11ax (Wi-Fi 6)
ATM Connectivity	MPLS / VPN
Banking Security	Next-Generation Firewall + IPS
Cabling	Cat6/Cat6A
Backbone	Optical Fiber

Interpretation:

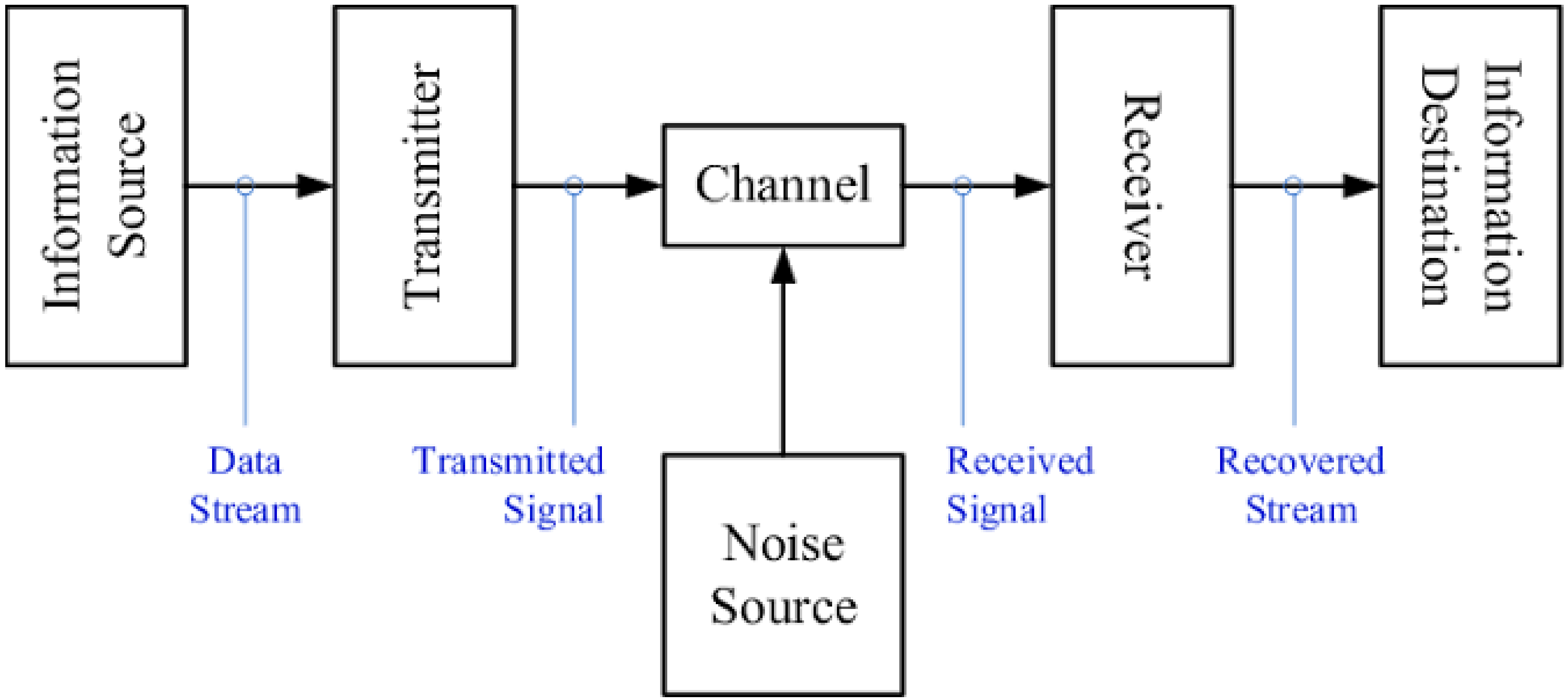
- Hence the network we have proposed uses an enterprise hierarchical design consisting of Router, Firewall, Core Switch, and Access Switches.
  - End devices such as employee computers, ATMs, printers, CCTV cameras, and Wi-Fi access points are connected through managed switches.
  - The router provides Internet and Head Office connectivity, while the firewall secures the banking infrastructure.
  - VLANs isolate sensitive services such as ATM transactions and guest Wi-Fi, ensuring confidentiality, integrity, and availability of banking services.

So this design offers high performance, scalability, fault tolerance, and strong security, making it suitable for a modern SBI smart branch.

Basic Communication System

- A basic communication system involves transmitting information from a source to a destination through a channel, involving components like a transmitter, channel, and receiver.

- The source generates the message, the transmitter prepares it for transmission, the channel provides the medium for signal propagation, and the receiver interprets the received signal.


Information Source
Transmitter
Channel
Receiver
Information Destination
Noise Source
Data Stream
Transmitted Signal
Received Signal
Recovered Stream

Components

- **Source:**
  - The origin of the information, such as a person speaking or a sensor reading.
- **Transmitter:**
  - Processes the information into a suitable signal for transmission over the chosen channel. This may involve encoding, modulation, and amplification.
- **Channel:**
  - The physical medium through which the signal travels. This can be wired (e.g., cables) or wireless (e.g., radio waves, light).
- **Receiver:**
  - The component that receives the transmitted signal and converts it back into a usable form. This may involve demodulation and decoding.
- **Destination:**
  - The intended recipient of the information.

Switching Techniques

Switching Techniques

- In computer networks, switching techniques determine how data is transmitted between devices.
- There are three primary methods: **circuit switching, message switching, and packet switching.**
- Each method offers different approaches to establishing connections, handling data, and optimizing network resources.

Circuit Switching

- A dedicated physical path is established between two communicating devices before any data transfer begins.
- Think of a telephone call where a physical connection is made between two phones before a conversation can occur.
- **Characteristics:**
  - Dedicated, reserved bandwidth for the duration of the communication.
  - Suitable for continuous, high-volume traffic.
  - Low latency once the connection is established.
  - Inefficient when traffic is intermittent.
- **Examples:** Traditional telephone networks, early data networks.

Message Switching

- Entire messages are sent from source to destination, passing through intermediate nodes that store and forward the message.
- **Characteristics:**
  - No dedicated path is established.
  - Each node stores the message completely before forwarding it.
  - Suitable for applications with variable message lengths and bursty traffic.
  - Potentially high latency due to storage and forwarding delays.
- **Examples:** Older email systems, some specialized military or scientific networks.

Packet Switching

- Data is divided into smaller packets, each routed independently through the network.
- Think of sending letters, where each letter is placed in an envelope and sent separately.
- **Characteristics:**
  - Efficient use of network resources, as bandwidth is shared dynamically.
  - Packets can take different routes to reach the destination.
  - Suitable for internet communication and most modern networks.
  - Can experience variable latency due to different packet routes.
- **Examples**: The internet, local area networks (LANs).

Why Internet Uses Packet Switching?

The Internet is built on **Packet Switching** because it is more efficient, reliable and scalable compared to Circuit Switching.

Advantages

No Dedicated Path
Packets from the same source to the same destination can take different paths.
No resources are reserved in advance.

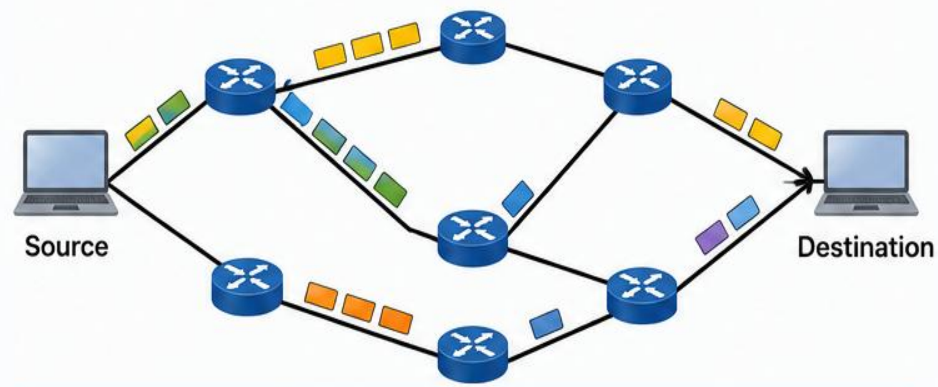
Better Bandwidth Utilization
Bandwidth is used only when packets are being transmitted.
No bandwidth is wasted when there is no data.

Fault Tolerant
If a link or node fails, packets can be routed through alternate paths.
Communication continues without interruption.

Dynamic Routing
Routers can choose the best path for each packet based on current network conditions (traffic, congestion, link status).

Scalable
Packet Switching supports a large number of users and devices efficiently.
Ideal for the growth of the Internet.

How Packet Switching Works (Simplified)



Packets may take different routes and arrive at different times, but are reassembled in the correct order at the destination.

Real-World Examples

- Web Browsing
- YouTube Streaming
- Video Calls (Google Meet)
- WhatsApp Messaging
- Cloud Services (Gmail, Drive)

Packet Switching is the foundation of the Internet because it delivers efficient, reliable and scalable communication.

Why Internet Uses Packet Switching?

- The Internet is built on Packet Switching because it is efficient, reliable, scalable, and fault tolerant.
- Instead of reserving a dedicated communication path, data is divided into packets that travel independently through the network and are reassembled at the destination.

- Advantages

1. No Dedicated Path
    - No communication path is reserved before transmission.
    - Multiple users can share the same network simultaneously.
    - Resources are allocated only when data is transmitted.
    - Example: While watching YouTube, you can simultaneously browse websites and send WhatsApp messages using the same Internet connection.

2. Better Bandwidth Utilization
    2. Bandwidth is shared among multiple users.
    3. Network resources are used only when packets are transmitted.
    4. Reduces bandwidth wastage.
    5. Example: Internet Service Providers (Jio, Airtel, BSNL) serve millions of users over shared infrastructure.
3. Fault Tolerant
    - If a router or communication link fails, packets are automatically rerouted through alternate paths.
    - Communication continues with minimal interruption.
    - Example: Banking transactions continue even when one ISP link fails because routers dynamically select another available path.

4. Dynamic Routing
   - Routers select the best available path based on network conditions.
   - Routing decisions consider congestion, delays, and link failures.
   - Example: Google and Amazon data centers use dynamic routing protocols to optimize network traffic.
5. Scalable
   - Supports millions of users and devices.
   - Easily accommodates growing Internet traffic without requiring dedicated circuits.
   - Example: Platforms such as YouTube, Netflix, WhatsApp, and Google Meet serve millions of concurrent users through packet-switched networks.

Packet Switching in Daily Life

Application	How Packet Switching is Used
WhatsApp Messaging	A text message is divided into packets and transmitted over the Internet. Even if packets follow different routes, the message is reconstructed correctly at the receiver.
YouTube Video Streaming	Video data is broken into thousands of packets and streamed continuously. Buffering allows smooth playback even if packets arrive at slightly different times.
Google Meet / Zoom	Voice and video are transmitted as packets in real time. Packet switching enables continuous communication with low latency.
Netflix	Movies are delivered as packets over Content Delivery Networks (CDNs). Adaptive streaming adjusts video quality according to available bandwidth.

Packet Switching in Daily Life

Application	How Packet Switching is Used
Gmail	Emails and attachments are divided into packets before transmission and reassembled at the recipient's mail server.
Online Banking (SBI, HDFC, ICICI)	Banking transactions travel as encrypted packets over secure Internet connections to ensure reliable and secure communication.
Cloud Services (Google Drive, OneDrive)	Files are uploaded and downloaded as packets, enabling efficient transfer of large amounts of data.

Why Not Circuit Switching?

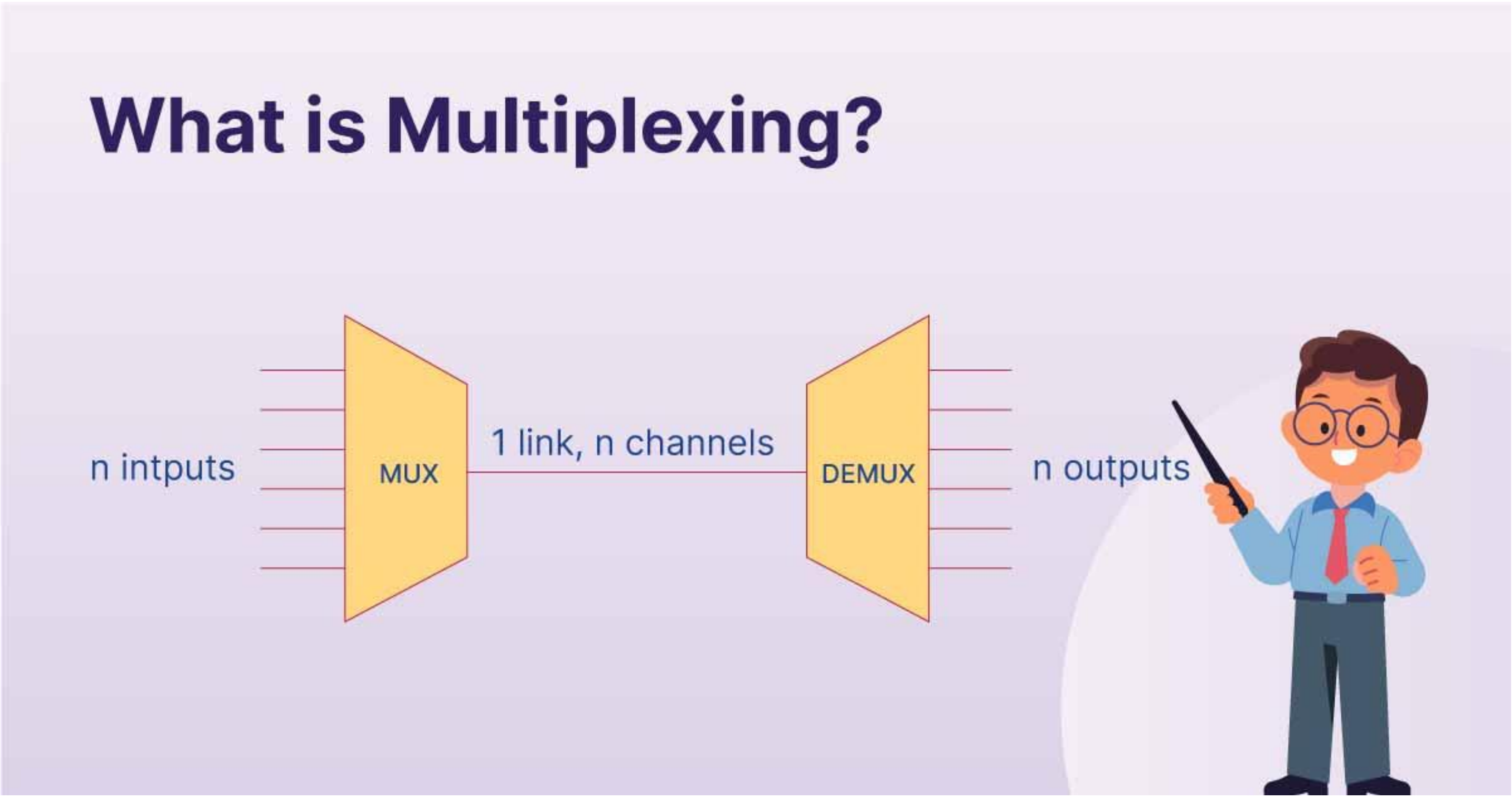
Circuit Switching	Packet Switching
Dedicated communication path	Shared communication path
Bandwidth remains reserved even when idle	Bandwidth is used only when data is transmitted
Suitable for traditional telephone systems	Suitable for Internet and data communication
Less efficient for bursty traffic	Highly efficient and scalable

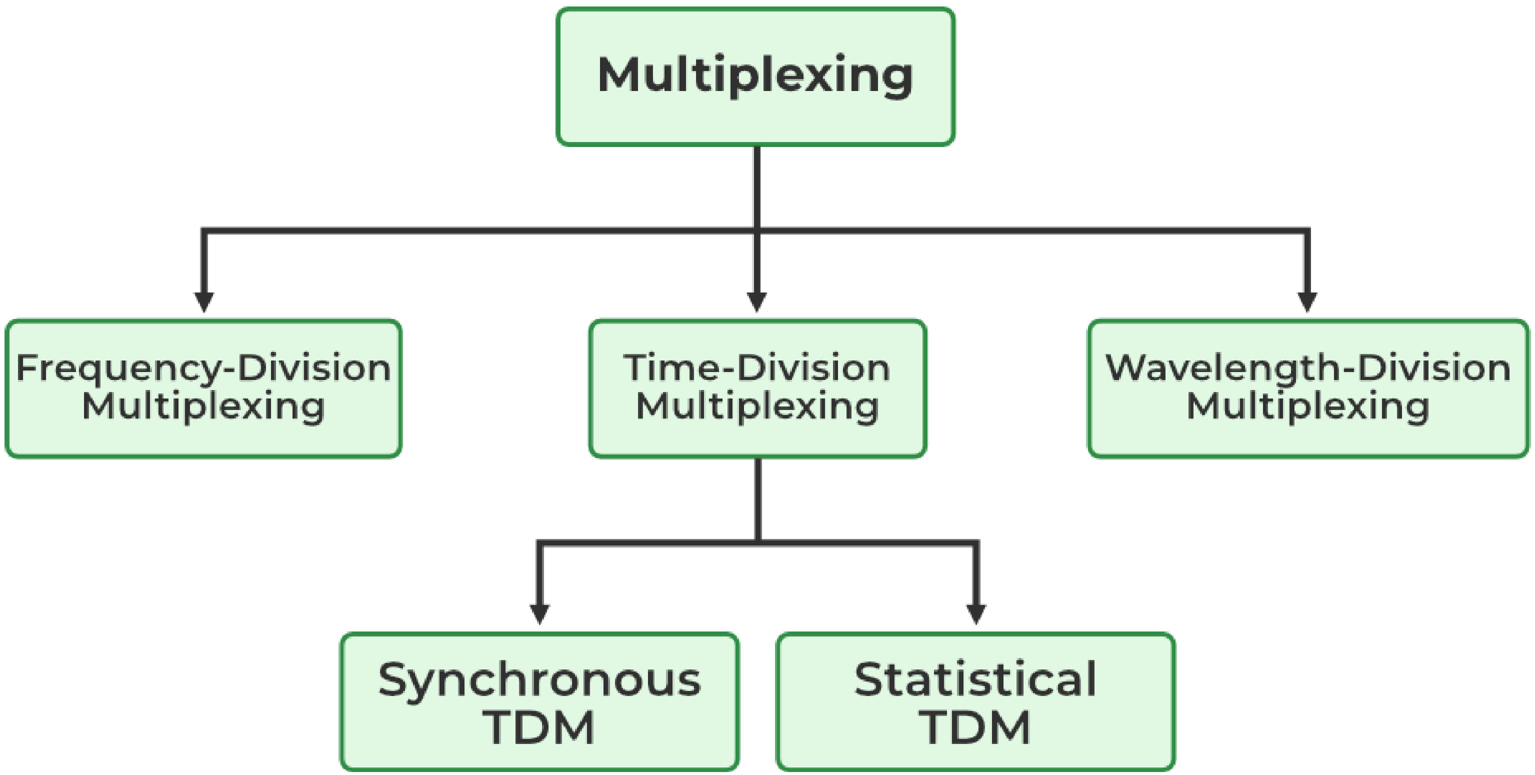
MULTIPLEXING

- Under the simplest conditions, a medium can carry only one signal at any moment in time.
- For multiple signals to share one medium, the medium must somehow be divided, giving each signal a portion of the total bandwidth.
- The current techniques that can accomplish this include frequency division multiplexing, time division multiplexing, and code division multiplexing.
- **A multiplexer** is a physical layer device that combines multiple data streams into one or more output channels at the source. Multiplexers multiplex the channels into multiple data streams at the remote end and thus maximize the use of the bandwidth of the physical medium by enabling it to be shared by multiple traffic sources.

Multiplexing

- Definition: Multiplexing is a technique used to transmit multiple signals simultaneously over a single communication channel, thereby improving bandwidth utilization and reducing transmission cost.
- Why Multiplexing is Needed
  - Efficient use of available bandwidth.
  - Reduces infrastructure cost by sharing one transmission medium.
  - Increases communication capacity.
  - Supports multiple users simultaneously.
  - Widely used in telephone systems, television broadcasting, satellite communication, fiber optics, and mobile networks.


What is Multiplexing?
n intputs
MUX
1 link, n channels
DEMUX
n outputs

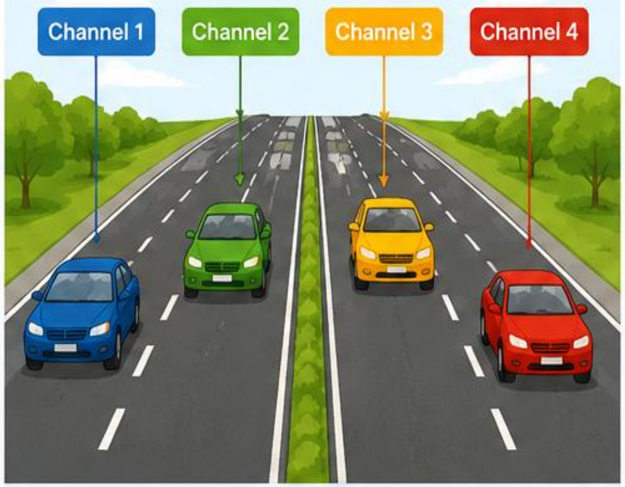

Multiplexing
Frequency-Division Multiplexing
Time-Division Multiplexing
Wavelength-Division Multiplexing
Synchronous TDM
Statistical TDM

Multiplexing – Road Analogy

Different data signals share the same communication medium using different techniques.

FDM
(Frequency Division Multiplexing)

Separate lanes



- The road has multiple separate lanes.
- Each lane is dedicated to one type of vehicle.
- Similarly, in FDM, each signal gets a separate frequency band.
- No interference between signals.

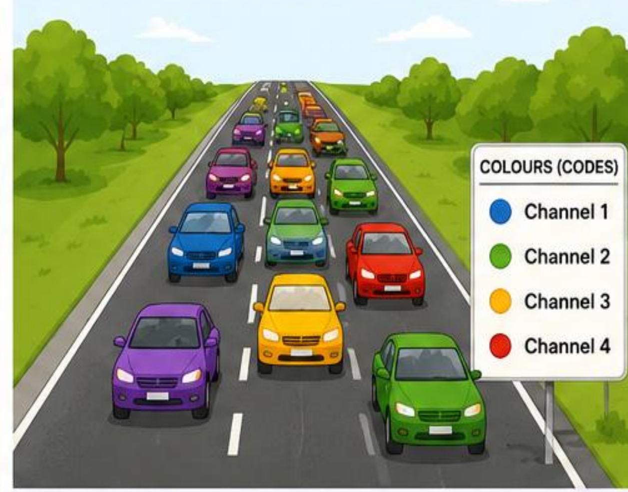
TDM
(Time Division Multiplexing)

One lane – Vehicles move one after another

- The road has only one lane.
- Vehicles use the lane one at a time in different time slots.
- Similarly, in TDM, all signals share the same frequency but take turns in time slots.
- Efficient use of bandwidth.

CDM
(Code Division Multiplexing)

Same road – Different coloured vehicles recognized by colour



- All vehicles use the same road at the same time.
- Each vehicle is identified by its colour.
- Similarly, in CDM, all signals share the same frequency and time but use different codes.
- Signals are separated at the receiver using the codes.

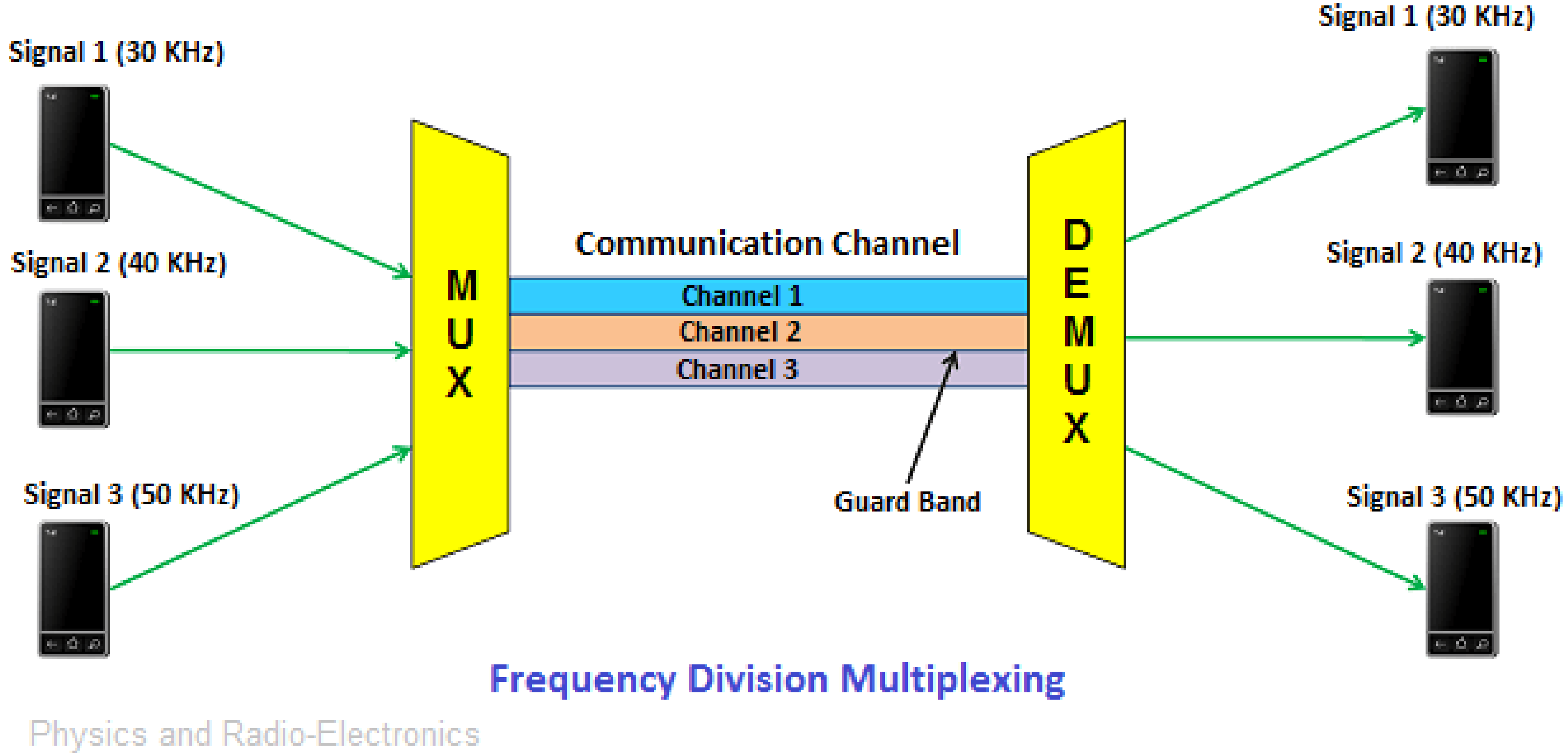
Key Idea: FDM separates by frequency, TDM separates by time, and CDM separates by code.

Frequency Division Multiplexing

- Frequency Division Multiplexing (FDM) is a multiplexing technique in which the available bandwidth of a communication channel is divided into multiple non-overlapping frequency bands, and each user is assigned a separate frequency band.
- All users transmit simultaneously, but at different frequencies.

Working Principle

- The communication channel has a fixed bandwidth.
- This bandwidth is divided into several frequency channels.
- Each input signal modulates a different carrier frequency.
- Guard bands are placed between adjacent channels to prevent interference.
- At the receiver, filters separate each frequency band.


Signal 1 (30 KHz)
Signal 2 (40 KHz)
Signal 3 (50 KHz)
M U X
Communication Channel
Channel 1
Channel 2
Channel 3
Guard Band
D E M U X
Signal 1 (30 KHz)
Signal 2 (40 KHz)
Signal 3 (50 KHz)
Frequency Division Multiplexing
Physics and Radio-Electronics

- **Characteristics**
  - Users transmit simultaneously.
  - Each user gets a unique frequency.
  - Requires guard bands.
  - Suitable for analog signals.
  - Continuous transmission.

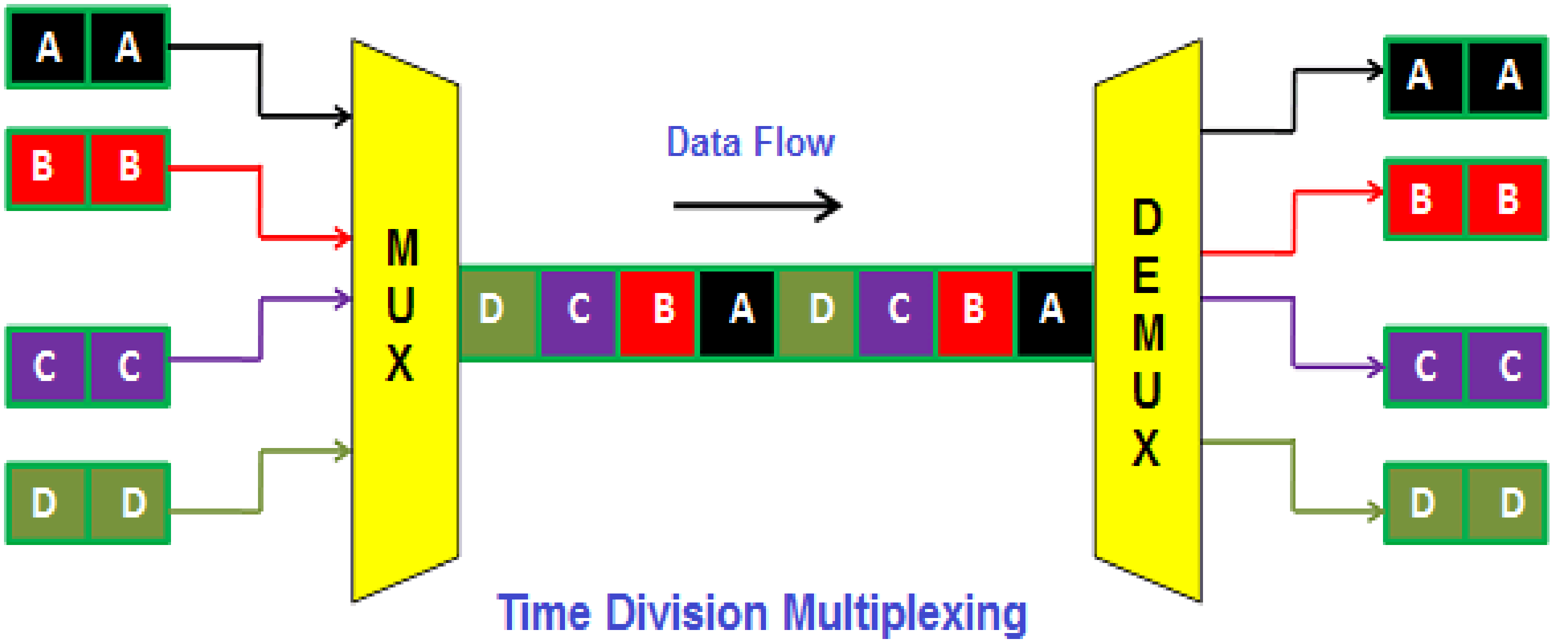
- **Advantages**
  - No synchronization required.
  - Continuous transmission.
  - Low delay.
  - Suitable for analog communication.
  - Multiple users can communicate simultaneously.

- **Disadvantages**
  - Guard bands waste bandwidth.
  - Cross-talk may occur.
  - More expensive hardware.
  - Fixed bandwidth allocation.
  - Not suitable for bursty traffic.

- **Applications**
  - FM Radio Broadcasting
  - Cable TV
  - Satellite Communication
  - Analog Telephone Systems
  - DSL Internet

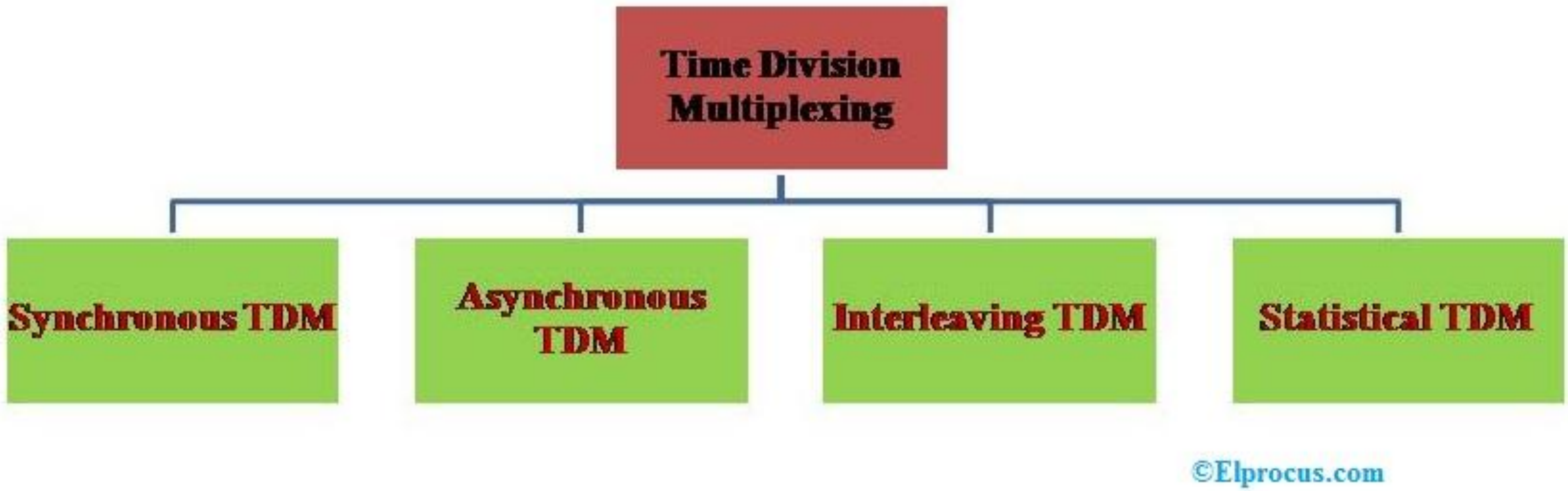
TIME DIVISION MULTIPLEXING

- Time Division Multiplexing is a technique where multiple users share the same communication channel by taking turns in different time slots.
- Each user gets the entire bandwidth, but only during its assigned time slot.
- Working Principle
  - Time is divided into equal slots.
  - Each user is assigned one slot.
  - Data is transmitted only during that slot.
  - The receiver reconstructs the original signals using synchronization.


A A
B B
C C
D D
MUX
Data Flow
D C B A D C B A
DEMUX
A A
B B
C C
D D
Time Division Multiplexing

- Characteristics
  - Same frequency shared.
  - Different time slots.
  - Requires synchronization.
  - Mostly used for digital communication.
- Advantages
  - Efficient for digital systems.
  - Better bandwidth utilization.
  - No guard bands.
  - Less interference.
  - Easier error detection.

- Disadvantages
  - Requires synchronization.
  - Delay increases with more users.
  - Empty slots waste bandwidth (Synchronous TDM).
  - Complex implementation.
- Applications
  - Digital Telephone Networks
  - ISDN
  - GSM Networks
  - SONET/SDH
  - Digital Multiplexers


Time Division Multiplexing
Synchronous TDM
Asynchronous TDM
Interleaving TDM
Statistical TDM
©Elprocus.com

Synchronous TDM

- Synchronous TDM assigns fixed time slots to every user, whether the user has data to send or not.
- Each user gets one slot in every frame.
- Characteristics
  - Fixed slots
  - Fixed frame size
  - Synchronization required
  - Simple implementation
- Advantages
  - Easy to design
  - Predictable delay
  - Suitable for real-time communication
  - Constant transmission speed
  - No waiting for slot allocation

- Disadvantages
  - Wastes bandwidth
  - Empty slots cannot be reused
  - Inefficient for bursty traffic
  - Poor bandwidth utilization
- Applications
  - PCM Telephony
  - ISDN
  - SONET
  - E1/T1 Telephone Systems
  - Voice Communication

Traditional telephone exchanges reserve one time slot for every phone call even if the speaker is silent.

Asynchronous TDM

- Time slots are allocated only to active users.
- If a user has no data, its slot is assigned to another active user.
- ***Bandwidth is fully utilized.***

- **Characteristics**
  - Variable slots
  - Dynamic allocation
  - Better utilization
  - Address information required

- Advantages
  - High bandwidth utilization
  - No empty slots
  - Efficient for computer networks
  - Suitable for burst traffic
- Disadvantages
  - More complex
  - Needs addressing
  - Variable delay
  - Buffer memory required

- Applications
- Computer Networks
- Packet Switching
- Internet Traffic
- LAN Communication

Cloud servers allocate bandwidth only to users currently sending data.

Interleaving TDM

- Interleaving TDM divides a user's data into smaller blocks or bytes and interleaves them with blocks from other users before transmission.
- Instead of sending an entire message from one user at once, pieces of messages from multiple users are alternated.

Why Interleaving?

- Suppose burst noise occurs.
- Without interleaving
- Entire message becomes corrupted.
- With interleaving
- Only small portions of every message are affected.
- Error correction becomes easier.

- Advantages
  - Reduces burst errors
  - Improves reliability
  - Better error correction
  - Better Quality of Service
- Disadvantages
  - More memory required
  - Processing delay
  - Complex implementation
- Applications
  - Satellite Communication
  - Deep Space Communication
  - Wireless Communication
  - Error Control Coding
  - Multimedia Streaming

4G/5G mobile systems use interleaving before channel coding to combat burst errors.

Statistical TDM

- Statistical TDM dynamically allocates time slots based on traffic statistics and user demand rather than assigning fixed slots.
- Only active users receive transmission opportunities.
- Characteristics
  - Dynamic slots
  - Intelligent allocation
  - High bandwidth efficiency
  - Traffic monitoring required

- Advantages
  - Highest bandwidth utilization
  - Supports burst traffic
  - Better Quality of Service
  - Less bandwidth wastage
  - Suitable for high-speed networks
- Disadvantages
  - High complexity
  - Expensive implementation
  - Prediction errors may cause delay
  - Requires intelligent scheduling algorithms
- Applications
  - Broadband Networks
  - Internet Backbone
  - Data Centers
  - Cloud Computing
  - High-Speed WAN

Internet Service Providers dynamically allocate bandwidth based on current user demand rather than giving every subscriber a fixed share at all times.

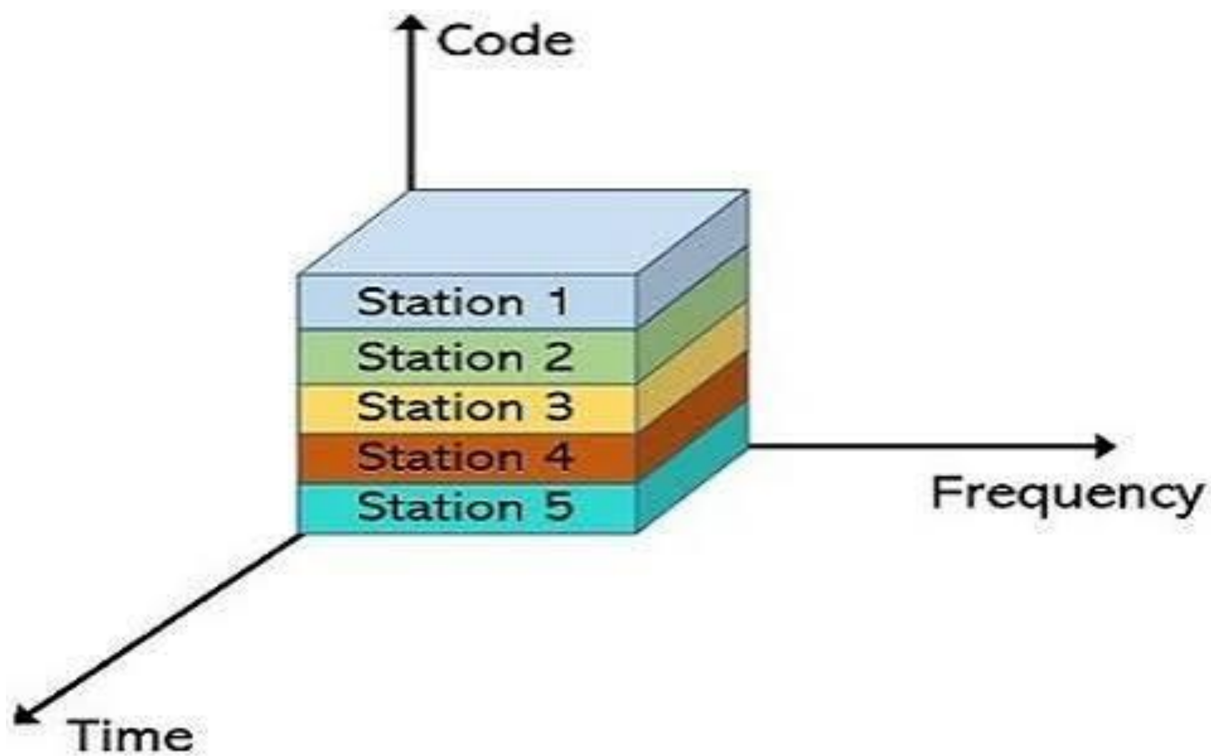
Feature	Synchronous TDM	Asynchronous TDM	Interleaving TDM	Statistical TDM
Slot Allocation	Fixed	Dynamic	Interleaved blocks	Dynamic based on traffic
Empty Slot	Yes	No	No	No
Bandwidth Utilization	Low	High	High	Very High
Complexity	Low	Medium	Medium–High	High
Delay	Constant	Variable	Slightly higher	Variable
Synchronization	Required	Required	Required	Required
Best For	Voice calls	Computer networks	Error-prone channels	Broadband and cloud networks
Example	T1/E1 Telephone	LANs	4G/5G, Satellite	ISP, Data Centers

Industry	TDM Type Used	Reason
Traditional Telephone Exchange	Synchronous TDM	Fixed voice channels with predictable timing
Corporate LAN	Asynchronous TDM	Only active computers transmit, improving efficiency
4G/5G Mobile Networks	Interleaving TDM	Protects data from burst errors during wireless transmission
Cloud Data Centers	Statistical TDM	Dynamically allocates bandwidth based on current traffic demand
Internet Service Providers (ISP)	Statistical TDM	Maximizes bandwidth utilization across many users
Satellite Communication	Interleaving TDM	Enhances reliability over noisy communication channels

Code Division Multiplexing

- Code Division Multiplexing is a multiplexing technique in which all users transmit simultaneously over the same frequency and at the same time, but each user uses a unique spreading code.
- The receiver identifies each signal using its corresponding code.

- Working Principle
  - Every user is assigned a unique binary code.
  - Data is multiplied by this code (spreading).
  - All encoded signals are transmitted together.
  - The receiver extracts the desired signal using the matching code.


Code
Station 1
Station 2
Station 3
Station 4
Station 5
Frequency
Time

- Characteristics
  - Same frequency.
  - Same time.
  - Different unique codes.
  - High security.
  - Resistant to interference.
- Advantages
  - Excellent bandwidth utilization.
  - High security.
  - Resistant to jamming.
  - No guard bands.
  - Soft handoff in mobile communication.

- Disadvantages
  - Complex implementation.
  - Requires accurate code synchronization.
  - Near-far problem.
  - Higher processing requirements.
- Applications
  - 3G Mobile Networks
  - GPS
  - Military Communication
  - Satellite Communication
  - Secure Wireless Systems
- Example
  - Three mobile users are talking simultaneously.
  - All use: Same frequency & Same time
  - Each mobile phone has a unique spreading code, allowing the base station to separate each user's signal correctly.

Parameter	FDM	TDM	CDM
Resource Shared	Frequency	Time	Code
Transmission	Simultaneous	Sequential (time slots)	Simultaneous
Bandwidth	Divided into frequency bands	Entire bandwidth per slot	Entire bandwidth shared
Guard Bands	Required	Not required	Not required
Synchronization	Not required	Required	Code synchronization required
Interference	Moderate	Low	Very Low
Security	Low	Moderate	High
Complexity	Low	Moderate	High
Signal Type	Mostly Analog	Mostly Digital	Digital
Efficiency	Moderate	High	Very High
Typical Applications	FM Radio, Cable TV	Digital Telephony, SONET/SDH	3G Networks, GPS, Military Communication

Which Multiplexing Technique is Used Today?

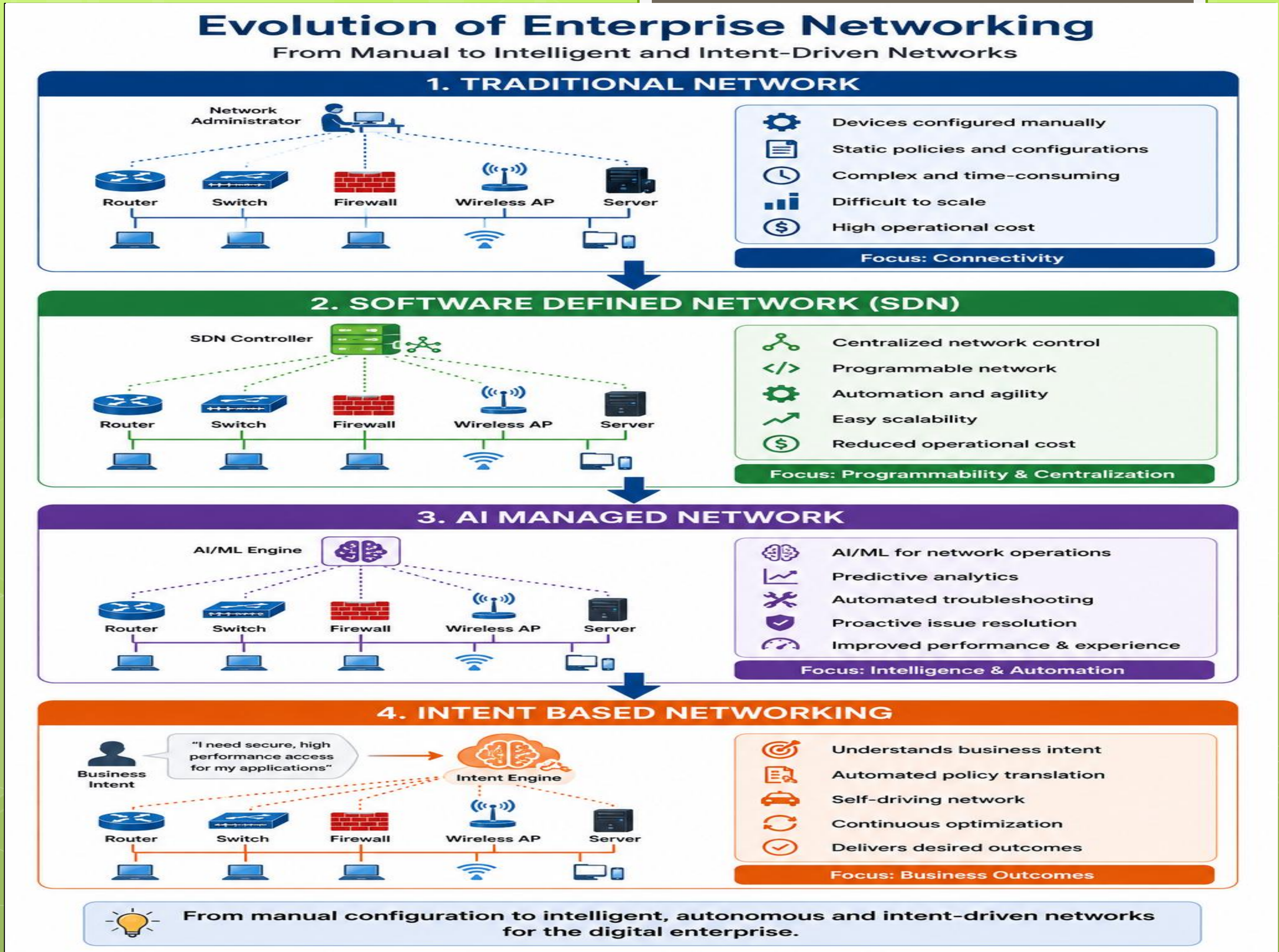
Technology	Multiplexing
FM Radio	FDM
Cable TV	FDM
3G	CDM
4G LTE	OFDMA (advanced FDM)
5G	OFDMA
Fiber Optics	WDM
Telephone	TDM

Modern Enterprise Networks

- Today's organizations use
  - SDN (Software Defined Networking)
  - SD-WAN
  - Wi-Fi 6E / Wi-Fi 7
  - Zero Trust Security
  - Network Automation
  - AI-based Traffic Monitoring
  - Cloud-managed Switches
- Example: Cisco Meraki, Juniper Mist AI, and Aruba Central

Companies Using Networking Technologies

Company	Technology
Google	SDN
Amazon AWS	SDN
Meta	AI Network Monitoring
Cisco	Meraki
Juniper	Mist AI
Airtel	MPLS + SD-WAN
Jio	Fiber + IPv6


Evolution of Enterprise Networking
From Manual to Intelligent and Intent-Driven Networks
1. TRADITIONAL NETWORK
Network Administrator
Router
Switch
Firewall
Wireless AP
Server
Devices configured manually
Static policies and configurations
Complex and time-consuming
Difficult to scale
High operational cost
Focus: Connectivity
2. SOFTWARE DEFINED NETWORK (SDN)
SDN Controller
Router
Switch
Firewall
Wireless AP
Server
Centralized network control
Programmable network
Automation and agility
Easy scalability
Reduced operational cost
Focus: Programmability & Centralization
3. AI MANAGED NETWORK
AI/ML Engine
Router
Switch
Firewall
Wireless AP
Server
AI/ML for network operations
Predictive analytics
Automated troubleshooting
Proactive issue resolution
Improved performance & experience
Focus: Intelligence & Automation
4. INTENT BASED NETWORKING
Business Intent
"I need secure, high performance access for my applications"
Intent Engine
Router
Switch
Firewall
Wireless AP
Server
Understands business intent
Automated policy translation
Self-driving network
Continuous optimization
Delivers desired outcomes
Focus: Business Outcomes
From manual configuration to intelligent, autonomous and intent-driven networks for the digital enterprise.